Component Selection

```
R = 1500;
C = 0.1e-6;

fc = 1/(2*pi*R*C)
```

```
fc =
1.0610e+03
```

Theoretical Frequency Response

```
f = logspace(log10(20), log10(10000), 500);

Hlo = 1 ./ sqrt(1+(2*pi*f*R*C).^2);
Hhi = (2*pi*f*R*C) ./ sqrt(1+(2*pi*f*R*C).^2);

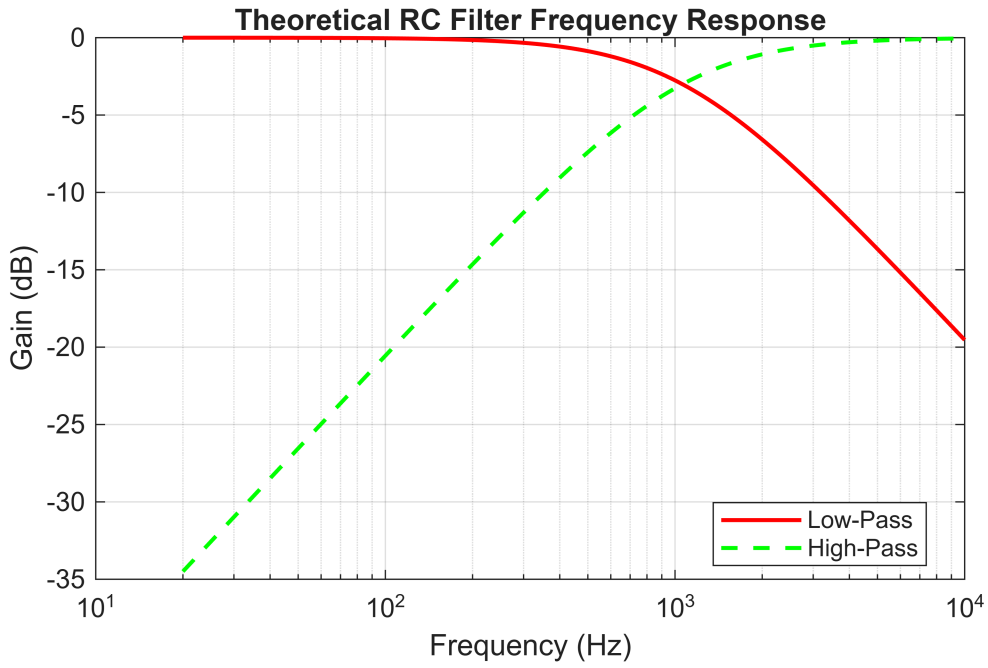
Gdblo = 20*log10(Hlo);
Gdbhi = 20*log10(Hhi);

figure

semilogx(f,Gdblo,'r-',f,Gdbhi,'g--','LineWidth',1.5)

xlabel('Frequency (Hz)')
ylabel('Gain (dB)')
title('Theoretical RC Filter Frequency Response')

legend('Low-Pass','High-Pass','Location','best')
grid on
```



Low-Pass Reference Measurements

```
Fs = 48000;
T = 2;

f_test = [100 250 500 1061 2000 5000 10000];

t = (0:1/Fs:T-1/Fs)';

Vin_lo = zeros(size(f_test));

for k = 1:length(f_test)

    x = 0.05*sin(2*pi*f_test(k)*t);

    player = audioplayer(x,Fs);
    rec = audiorecorder(Fs,16,2);

    record(rec);
    pause(0.3);

    playblocking(player);

    stop(rec);

    y = getaudiodata(rec);

    segment = y(round(0.7*Fs):round(1.3*Fs),1);

    Vin_lo(k) = sqrt(mean(segment.^2));
end
```

Low-Pass Experimental Measurements

```
Vout_lo = zeros(size(f_test));

for k = 1:length(f_test)

    x = 0.05*sin(2*pi*f_test(k)*t);

    player = audioplayer(x,Fs);
    rec = audiorecorder(Fs,16,2);

    record(rec);
    pause(0.3);

    playblocking(player);
```

```
    stop(rec);

    y = getaudiodata(rec);

    segment = y(round(0.7*Fs):round(1.3*Fs),1);

    Vout_lo(k) = sqrt(mean(segment.^2));

end
```

Low-Pass Theoretical vs Experimental Comparison

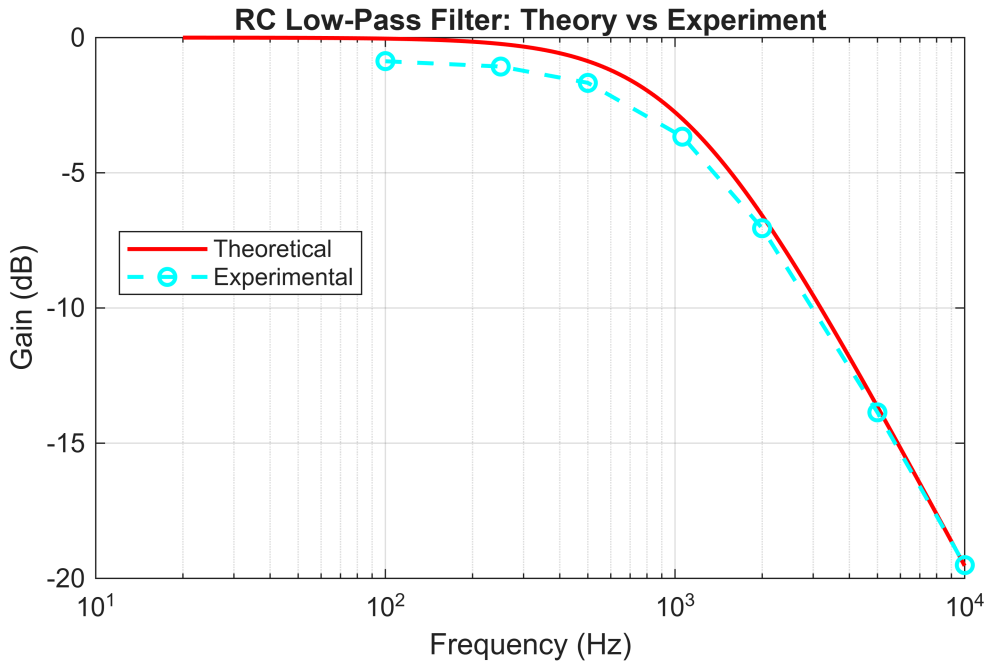
```
G_measured_lo = 20*log10(Vout_lo./Vin_lo);

figure

semilogx(f,Gdblo,'r-','LineWidth',1.5)
hold on
semilogx(f_test,G_measured_lo,'co--','LineWidth',1.5)

xlabel('Frequency (Hz)')
ylabel('Gain (dB)')
title('RC Low-Pass Filter: Theory vs Experiment')

legend('Theoretical','Experimental','Location','best')
grid on
```




```
% Calculate theoretical gain at measured frequencies
Hlo_test = 1 ./ sqrt(1+(2*pi*f_test*R*C).^2);

G_theoretical_lo = 20*log10(Hlo_test);
```

```
% Calculate error
Error_lo = G_measured_lo - G_theoretical_lo;

% Display results
results_lo = table(f_test',G_theoretical_lo', ...
    G_measured_lo',Error_lo', ...
    'VariableNames',{'Frequency_Hz','Theory_dB', ...
    'Measured_dB','Error_dB'});

disp(results_lo)
```

Frequency_Hz	Theory_dB	Measured_dB	Error_dB
100	-0.038407	-0.87519	-0.83678
250	-0.23465	-1.0707	-0.836
500	-0.87095	-1.673	-0.80207
1061	-3.0102	-3.665	-0.65482
2000	-6.583	-7.0524	-0.46934
5000	-13.656	-13.86	-0.20344
10000	-19.534	-19.507	0.027262

High-Pass Reference Measurements

```
Fs = 48000;
T = 2;

f_test = [100 250 500 1061 2000 5000 10000];

t = (0:1/Fs:T-1/Fs)';

Vin_hi = zeros(size(f_test));

for k = 1:length(f_test)

    x = 0.05*sin(2*pi*f_test(k)*t);

    player = audioplayer(x,Fs);
    rec = audiorecorder(Fs,16,2);

    record(rec);
    pause(0.3);

    playblocking(player);

    stop(rec);

    y = getaudiodata(rec);

    segment = y(round(0.7*Fs):round(1.3*Fs),1);
```

```
    Vin_hi(k) = sqrt(mean(segment.^2));

end
```

High-Pass Experimental Measurements

```
Vout_hi = zeros(size(f_test));

for k = 1:length(f_test)

    x = 0.05*sin(2*pi*f_test(k)*t);

    player = audioplayer(x,Fs);
    rec = audiorecorder(Fs,16,2);

    record(rec);
    pause(0.3);

    playblocking(player);

    stop(rec);

    y = getaudiodata(rec);

    segment = y(round(0.7*Fs):round(1.3*Fs),1);

    Vout_hi(k) = sqrt(mean(segment.^2));

end
```

High-Pass Theoretical vs Experimental Comparison

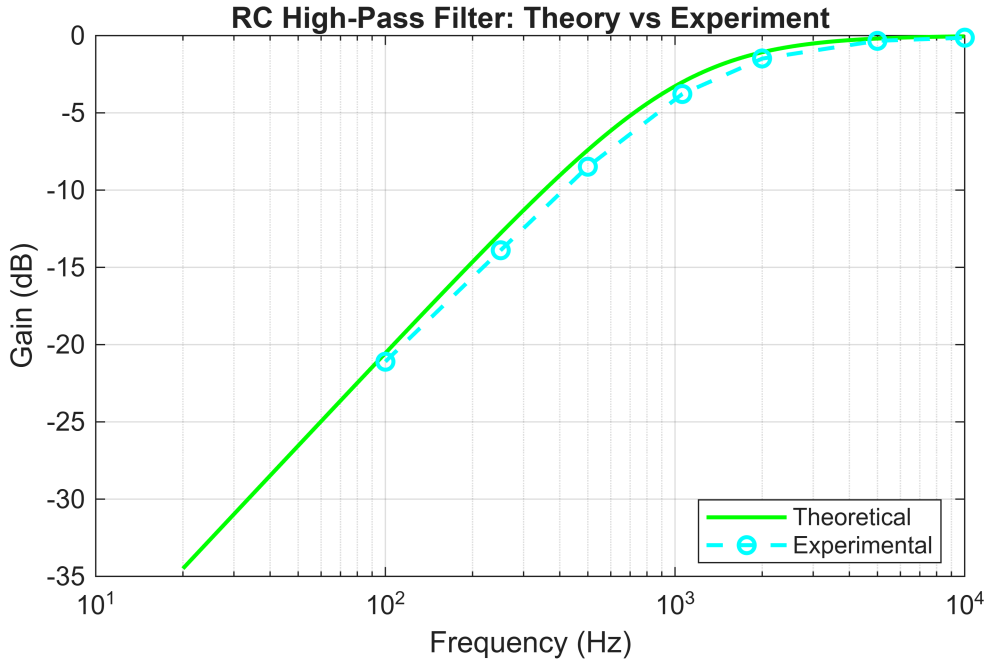
```
G_measured_hi = 20*log10(Vout_hi./Vin_hi);

figure

semilogx(f,Gdbhi,'g-','LineWidth',1.5)
hold on
semilogx(f_test,G_measured_hi,'co--','LineWidth',1.5)

xlabel('Frequency (Hz)')
ylabel('Gain (dB)')
title('RC High-Pass Filter: Theory vs Experiment')

legend('Theoretical','Experimental','Location','best')
grid on
```




```
results_hi = table(f_test',G_measured_hi', ...
    'VariableNames',{'Frequency_Hz','Measured_dB'});

disp(results_hi)
```

Frequency_Hz	Measured_dB
100	-21.098
250	-13.895
500	-8.4831
1061	-3.773
2000	-1.4888
5000	-0.33796
10000	-0.13922

High-Pass Error Analysis

```
Hhi_test = (2*pi*f_test*R*C) ./ ...
    sqrt(1+(2*pi*f_test*R*C).^2);

G_theoretical_hi = 20*log10(Hhi_test);

Error_hi = G_measured_hi - G_theoretical_hi;

results_hi = table(f_test',G_theoretical_hi', ...
    G_measured_hi',Error_hi', ...
    'VariableNames',{'Frequency_Hz','Theory_dB', ...
    'Measured_dB','Error_dB'});

disp(results_hi)
```

Frequency_Hz	Theory_dB	Measured_dB	Error_dB
100	-20.553	-21.098	-0.5446
250	-12.79	-13.895	-1.1042
500	-7.4061	-8.4831	-1.077
1061	-3.0104	-3.773	-0.76252
2000	-1.077	-1.4888	-0.41183
5000	-0.19129	-0.33796	-0.14667
10000	-0.048619	-0.13922	-0.090597